

# Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

Università di Pisa, A.A. 2025/2026, Corso di laurea magistrale in Fisica

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High temperature superconductivity in copper oxides (cuprates) represents one of the most striking phenomena in condensed matter physics. These materials host superconducting states with the highest known critical temperatures, emerging upon doping antiferromagnetic insulator compounds. The antiferromagnetic (AF) insulating and superconducting (SC) phases defy standard descriptions in terms of band picture and conventional phonon-mediated superconductivity. In fact, these phenomena are genuine signatures of strong electronic correlations which hinder effective single-particle descriptions of the many-electron system.

After 40 years since their discovery, it is widely accepted that the Hubbard model on the square lattice provides a minimal description of the low-energy electronic properties of cuprates. The model describes the competition between the electronic kinetic energy, modelled by nearest-neighbours hopping  $t$ , and the Coulomb repulsion, modelled by a purely local interaction potential  $U$ . At half-filling, this competition results in the so-called “Mott localisation” phenomenon, which distinguishes the metallic phase for  $U/t \ll 1$  from an insulating phase for  $U/t \gg 1$ . The latter is characterised by the almost complete localisation of electrons due to the high energetic cost of hopping towards occupied sites. Interestingly, this basic competition is intimately related to AF ordering and, upon doping, to the formation of the SC state with characteristic  $d$ -wave symmetry.

The fact that a purely repulsive interaction  $U$  is responsible for the formation of a SC state is remarkable, being superconductivity typically associated with an effective attractive interaction. In fact, a  $d$ -wave SC state can also be obtained, in general, by considering an attractive potential between electrons of neighbouring sites. Motivated by this observation, in this thesis, we study a modified version of the Hubbard model, dubbed as the “extended Hubbard model”, in which we include a nearest-neighbours attractive potential  $-V$ . The general goal of this work is to explore the interplay between the purely local repulsion and the non-local attraction. This is not only motivated by the fact that both interactions are, in principle, responsible for  $d$ -wave superconductivity, but also by recent experimental observations which strongly suggest that the term  $-V$  needs to be included in order to accurately describe the magnetic spectra of some specific compounds.

In this thesis, we study the extended Hubbard model using the Hartree-Fock method. Namely, we approximate the ground state by implementing a variational search in the subspace of the many-body Hilbert space of gaussian states. Despite its simplicity, the Hartree-Fock method provides a powerful framework to study competition among different phases, originating from different symmetry breaking mechanisms. In particular, we study the competition between three different phases: the normal phase, characterised by no symmetry breaking, the AF phase, characterised by a lowering of  $SU(2)$  symmetry, and the SC phase, characterised by the breaking of  $U(1)$  charge conservation.

This thesis work led to a number of original results, which can be summarised by the general observation that the non-local attraction  $-V$  acts cooperatively with the local repulsion  $U$  for the stabilisation of the various phases hosted by the Hubbard model.

For the normal phase, we find that the non-local attraction acts as a source of Mott localisation, leading to an effective reduction of the electronic bandwidth. In particular, for values of  $V$  larger than a critical value and close to half-filling, we find a strong Mott localisation characterised by the complete flattening of the electronic bands. Interestingly, this result shows an example of Mott localisation that can be described at the level of the Hartree-Fock approximation, whereas Mott localisation in the conventional Hubbard model usually requires more sophisticated approximation schemes.

For the AF phase, we restrict our analysis to the half-filled case. We find that the pairing-induced Mott localisation leads to a significant enhancement of the magnetisation, primarily originating from the local repulsion  $U$ .

Finally, we focus on the superconducting phase and derive self-consistency equations in both singlet and triplet Cooper pairing channels, showing how Cooper pairing can be mediated simultaneously by both the local repulsion and the non-local attraction. Therefore, we numerically investigate the singlet channel and find evidence of “symmetry mixing”, namely, the formation of a superconducting gap that is neither symmetric nor anti-symmetric under rotations of  $\pi/2$  in the plane. We show that symmetry mixing is connected to the emergence of a finite nematic order parameter, that signals the breaking of planar rotational symmetry. This nematic symmetry breaking results in the fact that the renormalisation of bands becomes anisotropic, producing a difference in electronic kinetics in the  $x$  and  $y$  directions.

To summarise, the original results obtained in this thesis show that the non-local pairing provides a different route towards the formation of key strongly correlated phenomena, including Mott localisation and nematic superconductivity, and call for future investigation using more sophisticated methods such as, for example, the Dynamical Mean Field Theory.